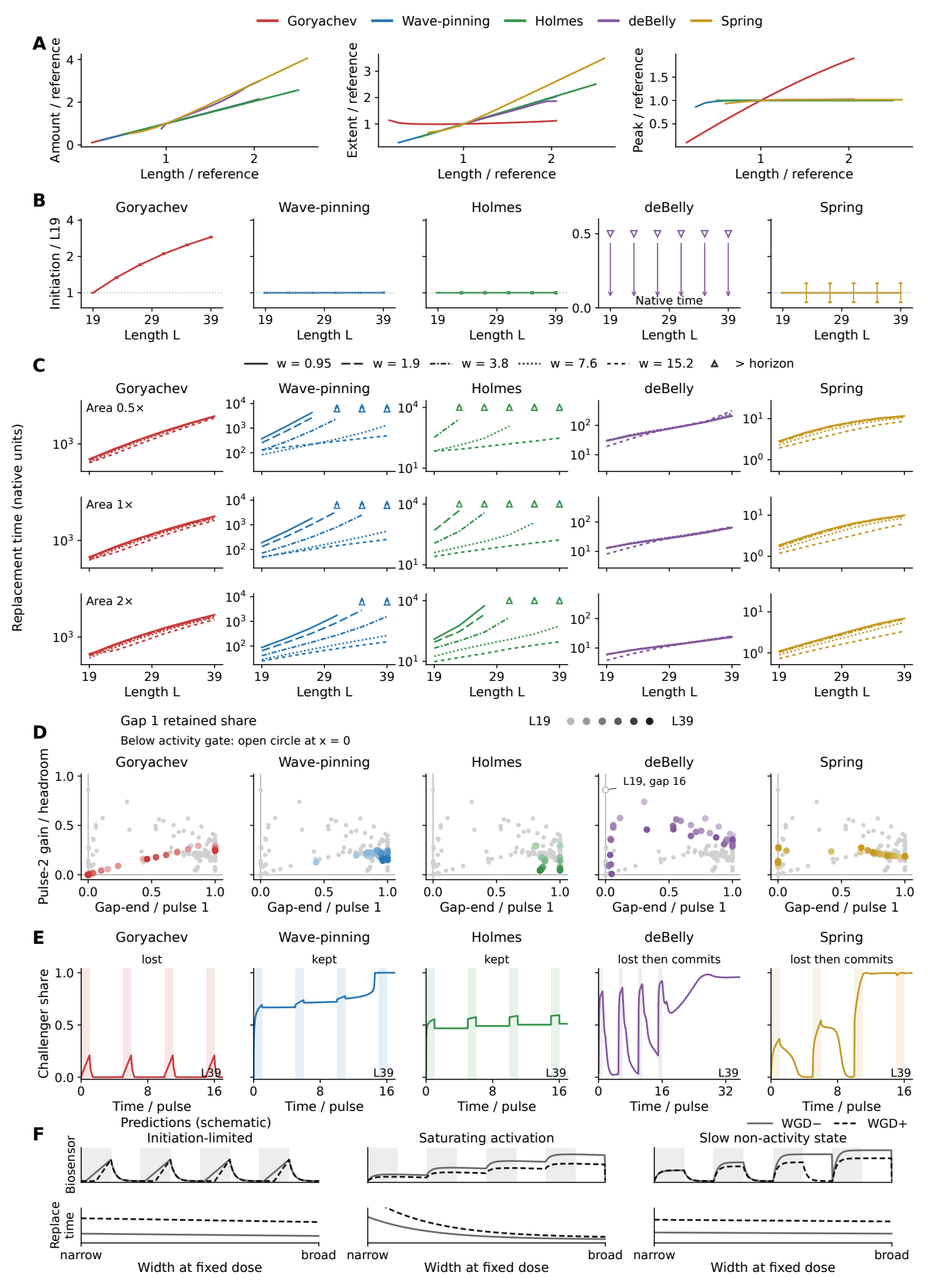

Goryachev
Wave-pinning
Holmes
deBelly
Spring
A
Amount / reference
Length / reference
Extent / reference
Peak / reference
B
Initiation / L19
Native time
Length L
C
w = 0.95
w = 1.9
w = 3.8
w = 7.6
w = 15.2
> horizon
Area 0.5×
Area 1×
Area 2×
Replacement time (native units)
D
Gap 1 retained share
Below activity gate: open circle at x = 0
L19
L39
L19, gap 16
Pulse-2 gain / headroom
Gap-end / pulse 1
E
lost
kept
lost then commits
Challenger share
Time / pulse
F
Predictions (schematic)
WGD−
WGD+
Initiation-limited
Saturating activation
Slow non-activity state
Biosensor
Replace time
narrow
broad
Width at fixed dose

A Activity profiles · Wave-pinning

cue support (fixed; also shown after cue-off) ◇ inter-region saddle | heading on arc axis

(i) C = 30, gap -1 straddling | (ii) C = 30, gap 4 separated | (iii) C = 47.6, gap -1 straddling | (iv) C = 47.6, gap 4 separated

Active species, u

0% | 50% | 90% | cue off

Arc position (native length)

B Time to 50% of cue-direction heading

Wave-pinning | Goryachev | Holmes | Spring

Time to 50%, t_{50} (native time)

C=47.6 | C=37.8 | C=30 | C=3.97 | C=3.15 | C=2.5

i | ii | iii | iv

grey band: heading undefined at ≥1 size

vertical line: first saddle at 50% (same C style)

△ time: lower bound

i–iv: profile placements in A

C Mean angular speed to 50%

Mean angular speed (rad / native time)

Cue edge gap (native length)

Cue edge gap (Spring native length)

grey band: heading undefined at ≥1 size

vertical line: first saddle at 50% (same C style)

△ time: lower bound

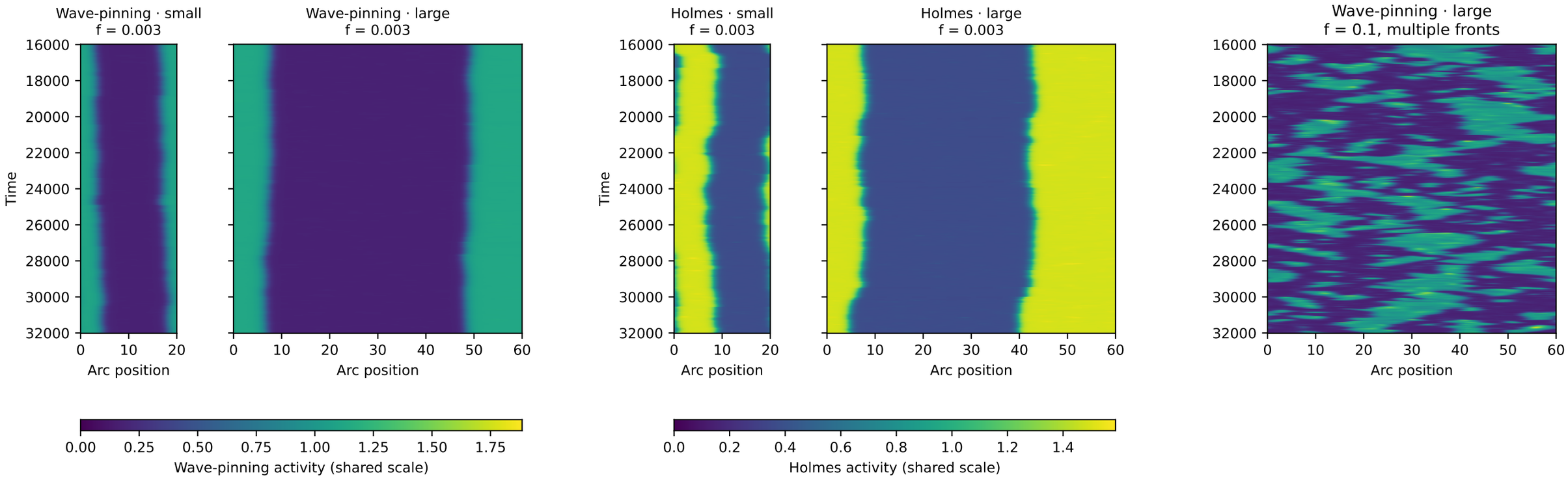
▽ speed: upper bound

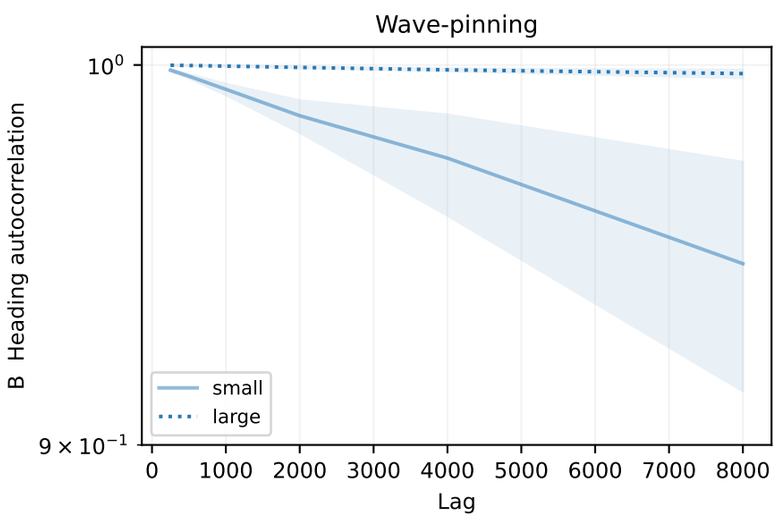
i–iv: profile placements in A

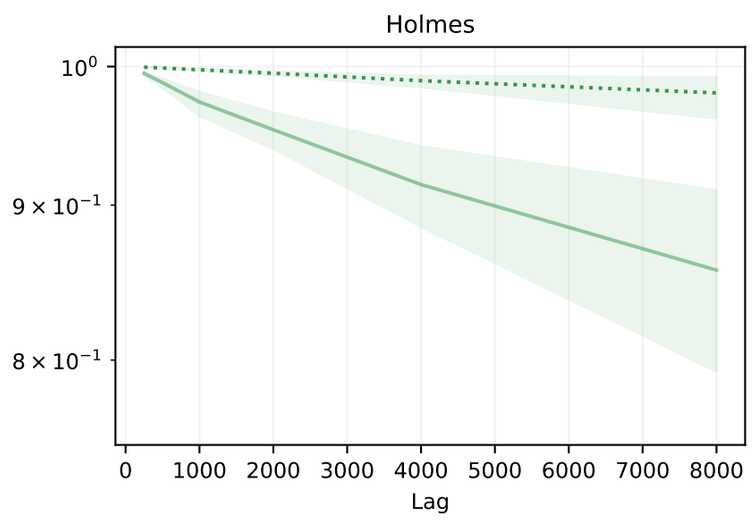
Enlarged cells hold heading longer under fluctuating input

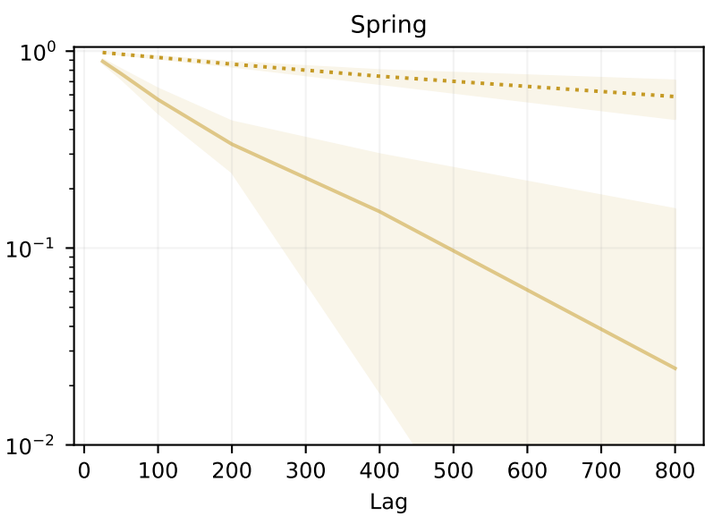
Gains reach or exceed geometry at each model's weakest measurable input; the advantage declines as input strengthens

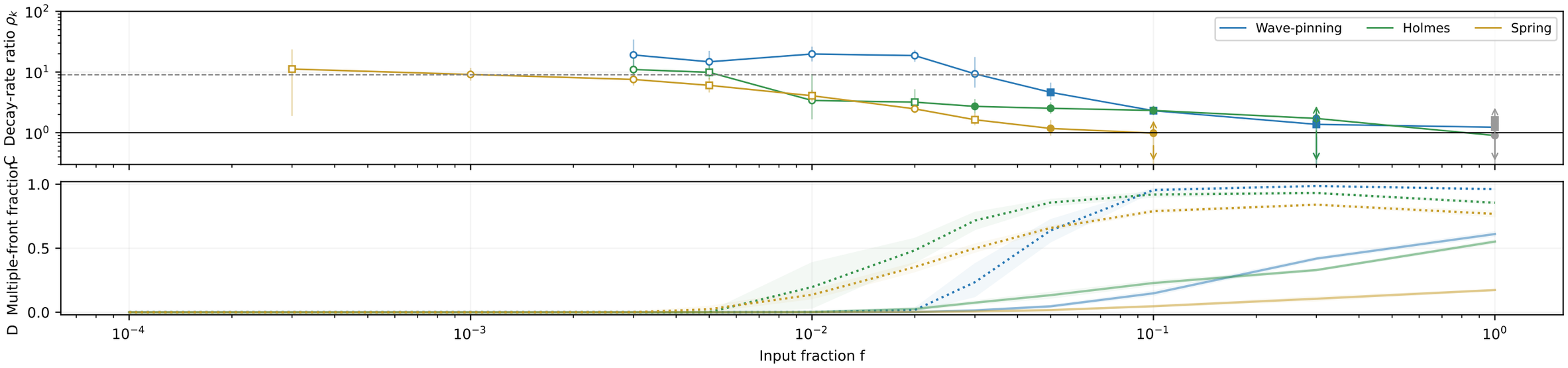
A Native activity histories

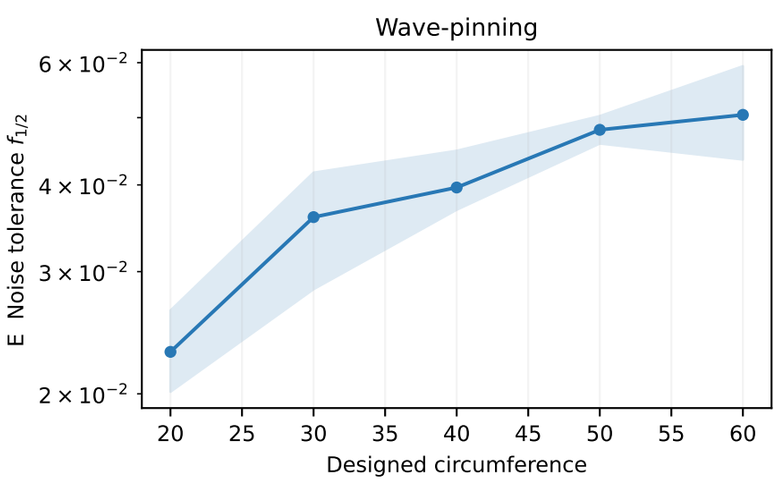


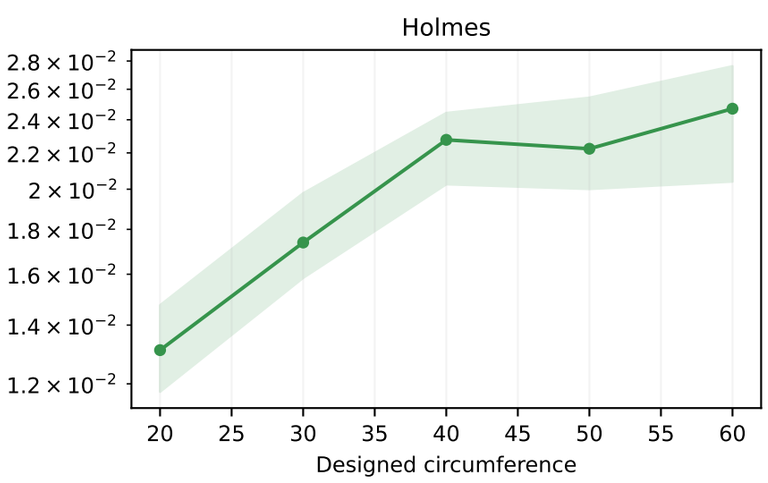


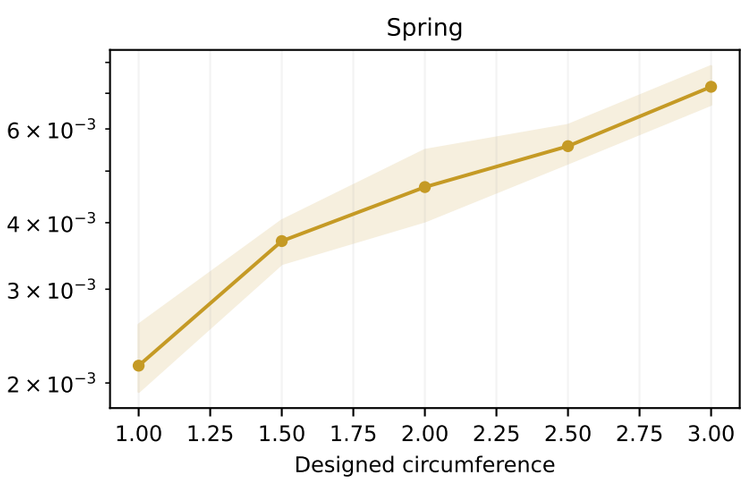












Small: lighter solid; large: darker dotted. C: open fill single front; filled multiple fronts; grey fill low contrast; square >1.5× lag drift.
Pointwise whole-seed 95% CIs. Horizontal references: ratio 1 and nominal geometry 9. D is descriptive; it does not establish a cause of erosion.
H1 uses a post hoc f < large-cell f½ domain; frozen H1 remains inconclusive. Goryachev retained in supplement only.